

Design Before You Build — Paper A

MEDIUM • 35 MINUTES • 30 MARKS

NAME

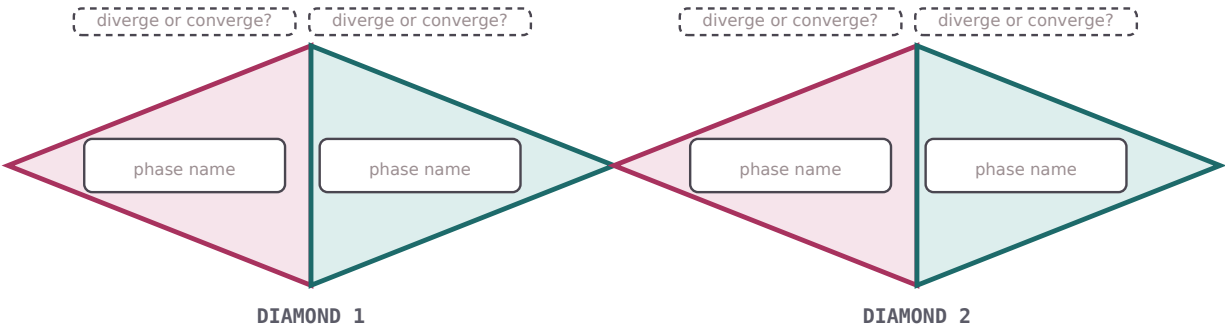
DATE

SCORE / 30

- Answer **every** question in the space provided. Show your working: method marks are awarded even when the final answer is wrong.
- No calculator and no computer. Every number on this paper works by hand.
- Where a question says **sketch**, draw plain boxes and labels only. Colour, real pictures and decoration earn nothing on a wireframe.
- Write **px** on every screen measurement. Give a contrast ratio as a number.
- The mark for each question is shown in square brackets on the right.
- Reminder: **contrast ratio** = $(L_1 + 0.05) \div (L_2 + 0.05)$, where L is relative luminance and L_1 belongs to the **lighter** colour. Ordinary text needs a ratio of at least **4.5**. Large text (24 px or bigger, or 18.66 px bold) needs at least **3**.

SECTION 1 – THE PROCESS AND THE WORDS

1. A school library is redesigning its book-renewal app and follows the double diamond. The diagram below has blank boxes.



- (a) Write the name of each phase in its box, in order, from left to right. [2]
- (b) In each dashed box above the diagram, write **diverge** or **converge**. [1]
- (c) State what the team is holding at the end of the first diamond. [1]

2. Each activity below comes from one phase of the library project. Write the phase in the right-hand box. Use each phase once. [4]

Activity	Phase
Watch six students try to renew a book at the desk	
Write one sentence: "Students return books late because they cannot see due dates"	
Sketch eight different ways to show a due date on the screen	
Release it to every student, then track late returns	

Two of these sound alike. Ask whether each one is about the problem or about the answer.

3. Each change below is made to the library app. Write **UI** or **UX**, and give a reason in one short sentence.

(a) Darken the pale grey due-date text so it can be read in sunlight. [1]

(b) Cut renewing a book from five taps to two. [1]

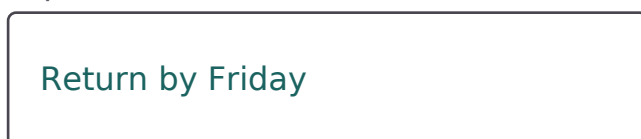
(c) Show "Renewed" for a moment after the tap. [1]

Did the screen change how it looks, or did the task change how it goes?

SECTION 2 – NUMBERS ON A SCREEN

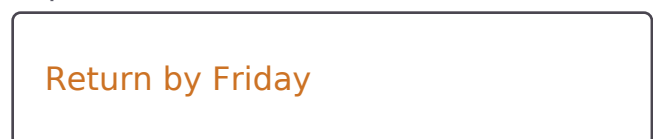
4. Two text colours are being tested on a white background for the due dates. L is relative luminance; white has $L = 1.00$.

Sample A



text #1B6360 L = 0.10
back #FFFFFF L = 1.00

Sample B



text #CC7224 L = 0.25
back #FFFFFF L = 1.00

(a) Calculate the contrast ratio for sample A. [1]

(b) Calculate the contrast ratio for sample B. [1]

(c) State which sample may be used for the ordinary 16 px due dates, and why. [1]

(d) The designer wants to use the other sample for a 28 px heading. State whether that is allowed, using a number from your answers. [1]

5. The Renew button is drawn **32 px wide and 28 px high**. A tap target should be at least **48 × 48 px**. Invisible padding, the same on every side, is added around the button.

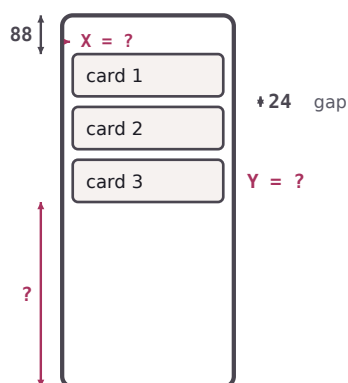
(a) Calculate the padding needed on each side to reach 48 px wide. [1]

(b) Calculate the padding needed on each side to reach 48 px high. [1]

(c) A designer adds only 4 px on every side. Calculate the new width and height, and state whether the button now meets 48 × 48. [1]

Padding goes on both sides of the button.

6. In Figma a designer makes a frame **390 px wide and 844 px high**. She places three cards in a column, each **342 px wide and 96 px high**. The top of card 1 is at **Y = 88**, there is a **24 px gap** between neighbouring cards, and the cards are centred from left to right.



FIGMA PANEL

Frame: W 390 H 844

All three cards: W 342 H 96

Gap between neighbouring cards: 24

Card 1: top edge at Y = 88

Cards are centred, left to right

X and Y are measured from the top-left corner of the frame. The ? marks are yours to find.

(a) State the X position of each card. [1]

(b) Calculate the Y position of card 3.

[2]

(c) Calculate the distance from the bottom of card 3 to the bottom of the frame.

[1]

SECTION 3 – JUDGING AND DRAWING

7. Look at the sign-in screen in Figure 1.

Shelf · school library

Sign in

Email

ananya.school.com

Password

.....

Log in

Sign up

CONTINUE AS GUEST

Figure 1

The sign-in screen of the school library app.
Ananya tapped Log in. The email she typed has no @.

This is the only sign that something is wrong.

Three buttons, three jobs, one screen.

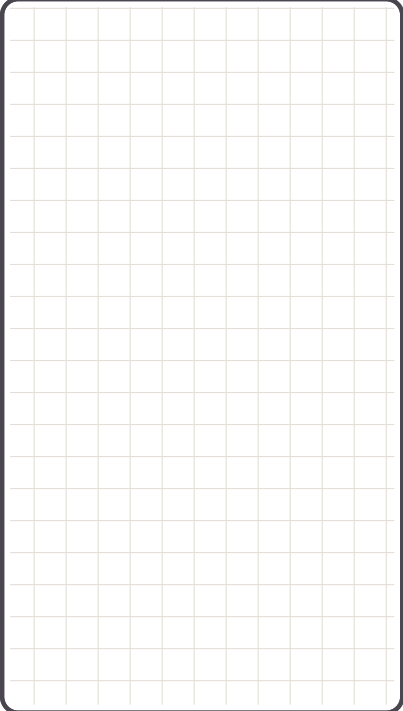
(a) Describe two different ways in which the buttons are inconsistent.

[2]

(b) The only sign that Ananya's email is wrong is the red border. Give one reason this fails for some users, and one addition that fixes it.

[2]

8. The library also needs a Home screen. In the frame below, sketch a **low-fidelity wireframe** for it with: navigation, one heading, one button and a footer, in a sensible order. You may add an image box. Label every box, and use the notes lines to say what the button says. [4]

HOME	390 × 844 frame	NOTES
		

Plain boxes and labels. Colour on a first wireframe is the mistake.

End of paper. Check that every screen measurement carries px, and that every UI or UX answer says why, not just which.